

POITIERS-BIARD AP, France

WMO: 073350

Lat: 46.5939N

Lon: 0.3144E

Elev: 423

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.2	25.8	17.1	13.2	25.8	20.9	15.9	28.5	28.1	49.8	25.1	48.8	6.2	30	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	89.7	69.0	85.9	67.3	82.3	66.1	70.9	84.6	69.2	81.9	67.6	79.0	7.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.2	75.3	64.9	93.6	73.3	63.5	89.0	71.9	34.9	84.9	33.5	81.9	32.2	79.1	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.7	17.4	DB	17.7	95.7	5.0	3.2	14.0	98.0	11.1	99.9	8.3	101.7	4.6	104.1	(4)
(5)			WB	17.0	74.1	5.0	1.7	13.4	75.3	10.4	76.3	7.6	77.3	3.9	78.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	41.0	42.0	46.9	51.2	57.3	64.0	67.5	67.2	61.3	55.5	46.6	41.8		
	DBStd	11.45	7.24	6.99	5.98	5.84	5.91	6.42	5.71	5.80	5.39	6.50	6.55	7.38		
	HDD50	1133	284	229	133	53	7	0	0	0	0	25	139	263		
	HDD65	4487	743	643	562	416	246	95	38	39	137	298	552	718		
	CDD50	2443	5	6	35	88	235	419	543	532	338	196	36	10		
	CDD65	324	0	0	0	1	9	64	115	106	25	4	0	0		
	CDH74	3777	0	0	0	26	159	742	1274	1221	326	29	0	0		
	CDH80	1282	0	0	0	1	26	240	450	484	78	3	0	0		
(14) Wind	WSAvg	8.3	9.2	9.5	9.1	8.6	8.0	7.7	7.7	7.2	7.2	7.6	8.4	9.1		
(15) Precipitation	PrecAvg	28.0	2.6	2.2	2.2	2.0	2.8	1.9	1.8	2.0	2.1	2.6	3.0	2.6		
	PrecMax	39.0	5.8	5.9	4.4	5.9	6.0	5.3	4.1	6.7	5.5	7.4	6.7	5.7		
	PrecMin	18.5	0.4	0.2	0.1	0.2	0.5	0.0	0.4	0.4	0.3	0.2	0.4	0.3		
	PrecStd	4.6	1.3	1.3	1.1	1.3	1.4	1.1	1.0	1.5	1.3	1.7	1.7	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	62.6	70.6	77.2	83.4	92.9	93.9	96.8	86.9	77.5	65.9	58.3		
		MCWB	53.0	51.3	55.3	60.6	66.1	70.7	69.2	70.0	65.7	63.9	58.2	54.0		
	2%	DB	54.9	57.5	65.6	72.1	78.5	86.2	89.1	90.0	81.8	72.8	61.5	56.2		
		MCWB	52.2	50.1	53.4	58.0	63.1	67.7	68.6	69.3	64.5	62.3	56.6	53.1		
	5%	DB	53.0	54.1	61.4	67.7	74.3	82.0	84.8	85.5	77.6	68.8	58.7	54.2		
		MCWB	50.5	49.4	52.0	56.0	61.5	66.4	66.9	67.6	63.7	61.0	55.3	51.8		
	10%	DB	51.0	51.9	57.6	63.1	70.3	77.8	81.1	80.9	73.5	65.7	56.3	52.3		
		MCWB	48.8	48.6	50.6	53.4	59.9	64.8	65.7	65.9	61.7	59.6	53.4	50.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	53.9	57.2	62.3	68.0	73.1	72.2	73.3	69.3	65.8	60.0	55.6		
		MCDB	55.6	58.5	67.0	73.7	79.8	86.0	87.5	88.4	79.9	72.7	62.8	56.8		
	2%	WB	52.8	51.9	55.0	59.2	65.1	69.8	70.3	71.0	67.0	63.8	57.7	53.8		
		MCDB	54.4	55.0	62.1	69.7	75.3	82.6	84.4	84.6	77.4	69.5	60.4	55.5		
	5%	WB	50.8	50.4	53.4	56.7	62.7	67.8	68.6	69.2	64.8	62.0	55.6	52.1		
		MCDB	52.6	53.4	59.3	65.7	72.0	79.3	81.3	82.0	74.4	67.4	58.2	54.0		
	10%	WB	49.0	48.9	51.6	54.4	60.5	65.7	67.0	67.2	63.0	60.3	53.6	50.4		
		MCDB	50.7	51.7	56.7	61.7	68.7	75.5	78.5	78.0	71.0	64.9	56.2	52.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	14.0	17.4	19.0	19.6	21.0	21.7	22.2	21.1	16.6	13.0	11.3		
		MCDBR	11.7	17.8	24.3	26.7	26.5	27.0	28.5	30.5	28.2	20.8	15.3	12.2		
		MCWBR	9.4	12.1	14.5	14.6	13.8	12.0	11.1	11.6	12.9	11.7	10.7	9.6		
	5% WB	MCDBR	10.4	14.7	19.8	23.6	23.7	23.6	24.3	25.2	22.7	17.4	13.7	10.8		
		MCWBR	9.0	10.9	12.7	13.6	13.3	11.5	10.7	11.0	11.8	10.4	10.3	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.339	0.367	0.393	0.397	0.411	0.400	0.391	0.363	0.360	0.334	0.317		
	taud		2.451	2.393	2.313	2.241	2.283	2.280	2.305	2.350	2.407	2.425	2.444	2.440		
	Ebn at Noon		245	263	271	273	276	272	273	271	269	251	236	231		
	Edh at Noon		22	28	35	41	41	41	40	36	31	27	22	20		
(44) All-Sky Solar Radiation	RadAvg		359	637	1014	1426	1670	1902	1854	1638	1284	777	434	314		
	RadStd		39	80	121	170	145	180	115	140	80	63	48	47		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page